
Assessment Test: Software Tester (QA) — My Answers

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1. Test Case Design

I'm writing scenarios first (the "what to check") and then a few sample test cases (the "how to check it") for each area, covering positive, negative, and edge cases as asked.

A. Registration

Test Scenarios:

1. User can register with all valid details.
2. User cannot register with an already- used email.
3. User cannot register with missing mandatory fields.
4. User cannot register with an invalid email format.
5. Password meets minimum security rules (length, special character, etc.).
6. User cannot register with a weak/short password.
7. Confirm-password field matches password field.
8. Special characters / emojis in the name field are handled properly.
9. Extremely long input in name/email field is handled without crashing.
10. Duplicate registration attempt with same details in quick succession (double-click on submit).

Sample Test Cases:

TC ID	Title	Steps	Test Data	Expected Result	Type
TC_REG_01	Register with valid details	1. Open registration page 2. Fill all fields correctly 3. Click Register	Name: Rahul Sharma, Email: rahul@test.com, Pwd: Test@123	User account is created, confirmation shown, redirected to login/dashboard	Positive
TC_REG_02	Register with existing email	1. Use an email already registered 2. Fill rest of form 3. Submit	Email: rahul@test.com (already used)	Error: "Email already registered"	Negative
TC_REG_03	Register with blank mandatory fields	1. Leave Name/Email/Password blank 2. Click Register	All fields empty	Field-level "required" error shown, form not submitted	Negative
TC_REG_04	Register with invalid email format	1. Enter wrong format email 2. Submit	Email: rahulattest.com	Error: "Enter a valid email"	Negative
TC_REG_05	Password boundary check	1. Enter password of exactly minimum allowed length	e.g., 8 characters if 8 is the min	Registration succeeds	Edge
TC_REG_06	Password below minimum length	1. Enter a 1-2 character password	Pwd: "ab"	Error shown, registration blocked	Negative

B. Login

Test Scenarios:

1. Login with correct email and password.
2. Login fails with wrong password.
3. Login fails with unregistered email.
4. Login fails with blank fields.
5. Password field masks the input (shows dots, not plain text).
6. Account lockout / warning after multiple wrong attempts (if that's a feature).
7. Session stays active as expected, and expires/logs out correctly.
8. "Forgot

password” flow works (if present). 9. SQL injection or script input in login fields is safely rejected. 10. Login with extra spaces before/after email (should still work after trimming).

Sample Test Cases:

TC ID	Title	Steps	Test Data	Expected Result	Type
TC_LOG_01	Login with valid credentials	1. Enter correct email & password 2. Click Login	Registered user	User lands on dashboard/task list	Positive
TC_LOG_02	Login with wrong password	1. Enter correct email, wrong password	rahul@test.com / wrongpass	Error: “Invalid credentials”, user stays on login page	Negative
TC_LOG_03	Login with unregistered email	1. Enter email not in system	notexist@test.com	Error: “User not found” / generic invalid credentials message	Negative
TC_LOG_04	Login with empty fields	1. Leave both fields blank 2. Click Login	Blank	Validation error, no request sent to server	Negative
TC_LOG_05	SQL injection attempt	1. Enter ‘ OR ‘1’=’1 in email/password	‘ OR ‘1’=’1	Login rejected, no bypass, no server error	Edge/Security
TC_LOG_06	Login with email having leading/trailing spaces	1. Enter “ rahul@test.com ” with spaces	Space + valid email	Login succeeds (spaces trimmed)	Edge

C. Task CRUD Operations

Test Scenarios: 1. Create a task with valid data (title, description, due date, etc.). 2. Cannot create a task with empty required fields (e.g., title). 3. View task list shows all tasks correctly for the logged-in user. 4. Task list updates immediately after create/edit/delete (no need to refresh). 5. Edit an existing task and confirm changes are saved. 6. Delete a task and confirm it’s removed from the list and database. 7. Cannot see/edit/delete another user’s tasks (data isolation). 8. Create task with very long title/description (boundary/edge case). 9. Pagination or scrolling works correctly when there are many tasks. 10. Deleting a task shows a confirmation prompt before permanent deletion.

Sample Test Cases:

TC ID	Title	Steps	Test Data	Expected Result	Type
TC_TSK_01	Create task with valid data	1. Login 2. Click “Add Task” 3. Fill title & details 4. Save	Title: “Buy groceries”	Task saved and visible in list	Positive
TC_TSK_02	Create task with empty title	1. Click “Add Task” 2. Leave title blank 3. Save	Title: “”	Error shown, task not created	Negative
TC_TSK_03	View task list	1. Login as user with existing tasks	User with 5 tasks	All 5 tasks displayed correctly with correct details	Positive
TC_TSK_04	Edit a task	1. Select a task 2. Change title 3. Save	Old: “Buy milk” → New: “Buy bread”	Updated title reflected in list and DB	Positive
TC_TSK_05	Delete a task	1. Select task 2. Click delete 3. Confirm	Any existing task	Task removed from list, not retrievable after refresh	Positive
TC_TSK_06	User A cannot see User B’s tasks	1. Login as User A 2. Check task list	Two separate accounts	Only User A’s own tasks are visible	Negative/Security
TC_TSK_07	Create task with extremely long title	1. Enter a 1000+ character title 2.	Very long string	Either handled gracefully with a	Edge

		Save		max-length limit, or saved without breaking UI	
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D. Input Validation & Error Handling

Test Scenarios: 1. All mandatory fields show proper validation messages when left empty. 2. Special characters (<, >, ', ", script tags) don't break the app or cause XSS. 3. Numeric fields (if any, like due date) reject non-numeric/invalid input. 4. Date fields don't accept past dates where not logical (e.g., due date in the past, if disallowed). 5. Error messages are clear, specific, and don't expose technical/system details. 6. App behaves correctly on network failure (e.g., submitting a task while offline). 7. File size/type validation if attachments are allowed with tasks. 8. Form doesn't allow double-submission on slow network (no duplicate task created on double-click).

2. Bug Identification (8-10 Potential Bugs/Risk Areas — Without Executing the App)

Since I haven't run the app, I'm listing these purely based on the requirements shared, thinking about common places where task-management apps usually go wrong.

#	Bug/Risk Area	Severity	Reason/Impact
1	No mention of password complexity rules during registration	Major	Weak passwords could be accepted, leading to easily compromised accounts
2	Possible lack of data isolation between users' tasks	Critical	If not properly implemented, one user could view/edit/delete another user's tasks — major privacy/security issue
3	No confirmation dialog before task deletion	Major	Accidental clicks could permanently delete tasks with no way to undo
4	Missing duplicate-email check during registration	Major	Could allow multiple accounts with the same email, causing login/identity conflicts
5	No input sanitization mentioned for task title/description	Critical	Could open the app to XSS or SQL injection if special characters aren't handled
6	Task list may not refresh automatically after create/edit/delete	Minor	User might think the action failed and retry, causing duplicate entries
7	No mention of session timeout/expiry handling	Major	Sessions staying open indefinitely could be a security risk on shared devices
8	No field length limits specified for title/description	Minor	Extremely long text could break UI layout or cause performance issues
9	No error handling mentioned for network/server failure during CRUD actions	Major	User could lose data or see unclear errors if the request fails mid-way
10	No mention of what happens to tasks if a user account is deleted	Minor	Orphaned tasks could remain in the database, wasting storage / causing confusion